

CTD

CTD is a cancer drug treatment. It is used to treat myeloma.

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What is CTD?

CTD is a combination of cancer drugs. It is used to treat [myeloma](#).

CTD is named after the initials of the drugs used in the treatment. The drugs are:

- C – [cyclophosphamide](#) (a chemotherapy drug)
- T – [thalidomide](#) (an immunotherapy drug)
- D – dexamethasone (a [steroid](#)).

It is best to read this information with our general information about [cancer drug treatments](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How CTD is given

CTD comes as capsules and tablets. This means you can take it at home.

During your course of treatment, you will meet with someone from your cancer team, such as a:

- cancer doctor
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your cancer drugs.

Your course of treatment

You usually have a course of several cycles of treatment over a few months. Your cancer team will discuss your treatment plan with you. They may give you a copy of the treatment plan to take home.

A cycle of CTD is usually given over 21 days (3 weeks) or 28 days (4 weeks). Your doctor, nurse or pharmacist will tell you which cycle you are going to have. You might take:

- cyclophosphamide once a week for either 3 or 4 weeks
- thalidomide every day for either 21 or 28 days
- dexamethasone as 2 short courses over 21 or 28 days.

Your cancer team will give you the capsules and tablets to take home. They may have different strengths. Always take them exactly as explained. This is important to make sure they work as well as possible for you.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take all your capsules or tablets exactly as they tell you to.

Taking CTD

Make sure you:

- swallow your tablets and capsules whole with a glass of water
- do not chew, open, break or crush them
- take them at the same time every day.

If you forget to take the cyclophosphamide or dexamethasone, take the missed dose as soon as possible within the same day. If a full day has passed, do not take a double dose. Contact the hospital on the 24-hour number you have been given for advice.

If you forget to take thalidomide and less than 12 hours have passed, take your capsules straight away. If more than 12 hours have passed, do not take your missed dose. Just take your usual dose at the usual time the next day. Tell your doctor, nurse or pharmacist if you have missed a dose.

Other things to remember about thalidomide and cyclophosphamide capsules:

- Wash your hands after taking your capsules.
- Do not touch the capsules if you are pregnant.
- Other people should avoid direct contact with them. If possible, you should handle the capsules yourself. If you need help, the person helping you should wear disposable gloves. If pregnant, the person should not touch or handle thalidomide at all.
- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.
- Do not throw away unused capsules or tablets. Return them to your cancer team at the hospital.
- Do not give blood for 7 days after taking thalidomide

It is important to:

- take dexamethasone with or just after breakfast
- take thalidomide at bedtime, as it can make you feel sleepy
- avoid grapefruit and grapefruit juice grapefruit juice can affect how cyclophosphamide works.

Preventing pregnancy while taking thalidomide

You must not become pregnant or make someone pregnant while taking thalidomide. This is because it may cause severe abnormalities in developing babies.

You will take part in a pregnancy prevention programme during treatment. Even if you do not think you are at risk of getting pregnant or making someone pregnant, you will be included in this programme.

Your doctor, nurse or pharmacist will give you information about the programme, and the risks of thalidomide and pregnancy. They will ask you to sign a consent form to show that you understand the information.

Pregnancy tests and contraception

If you are of childbearing age and able to get pregnant, you will have a pregnancy test before starting this treatment. The pregnancy test is repeated every 4 weeks during treatment, and 4 weeks after treatment finishes.

You must use an effective form of contraception, such as an implant, injection or the progesterone-only pill. You should not use the combined oral contraceptive pill. It increases your risk of developing blood clots.

You must use contraception to prevent pregnancy:

- for 4 weeks before treatment
- during treatment
- for 4 weeks after treatment finishes.

If you think you may be pregnant at any time during your treatment, contact your doctor, nurse or pharmacist straight away.

Contraception to prevent making someone pregnant

Thalidomide can pass into the semen that carries sperm. During treatment and for a week after treatment finishes, you must not:

- make someone pregnant
- donate sperm.

You can protect partners by using a condom during sex. During treatment and for a week after treatment finishes, you must use a condom during sex with anyone who is:

- pregnant
- of childbearing age and able to get pregnant.

If a partner thinks they might be pregnant during your treatment, contact your doctor, nurse or pharmacist straight away.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Feeling sick

You may feel sick during this treatment, but this is usually mild. Your doctor, nurse or pharmacist may prescribe you anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. [If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important

to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice.

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Hair loss

Your hair will get thinner. Or you may lose all the hair from your head. You may also lose your eyelashes and eyebrows, as well as other body hair. Hair loss usually starts after your first or second treatment.

If you want to cover up hair loss, there are different ways you can do this. Your nurse will give you information about [coping with hair loss](#).

Remember to [protect your skin from the sun](#). Use suncream with a sun protection factor (SPF) of at least 30 on your scalp. Or cover up with a hat or scarf.

Hair loss is almost always temporary. Your hair will usually grow back after treatment ends.

Increased appetite

[Steroids](#) can make you feel more hungry than usual, and you may [gain weight](#). Your appetite will go back to normal when you stop taking them. If you are worried about gaining weight, talk to your doctor or nurse.

Raised blood sugar levels

This treatment can raise your blood sugar levels and increase the risk of diabetes.

Signs of raised blood sugar include:

- feeling thirsty
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.

Tell your doctor or nurse if you have these symptoms. They can do tests to check your blood sugar level and give you advice.

If you already have [diabetes](#), your blood sugars may be higher than usual. You may need to check them more often, or make changes to your diabetic treatment. Your diabetic doctor or nurse will talk to you about how to manage this.

Mood and behaviour changes

Steroids can affect your mood. You may:

- feel anxious or restless
- have mood swings
- have problems sleeping.

Taking your steroids in the morning may help you sleep better at night.

Tell your doctor or nurse if you have any of these side effects. They may make some changes to your treatment if the side effects become a problem.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Dizziness, blurred vision or feeling drowsy

Thalidomide can cause dizziness, blurred vision or make you feel sleepy (drowsy). Try to sit upright for a few minutes before standing to help avoid dizziness.

Do not drink alcohol while you are taking thalidomide. This is because alcohol can make you sleepy and thalidomide can make you even sleepier. Steroids can also cause blurred vision.

Do not drive or operate machinery if you have these side effects. Talk to your doctor, nurse or pharmacist if you notice any of these effects.

Skin changes

This treatment can [affect your skin](#). It may cause a rash, which may be itchy. Your doctor or nurse can tell you what to expect. If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Always tell your doctor, nurse or pharmacist about any skin changes. They can give you advice and may prescribe creams or medicines to help. Changes to your skin usually improve when treatment ends.

Sometimes, thalidomide can cause a more serious skin rash that blisters. Do not take any more of this treatment and contact the hospital straight away on the 24-hour number if this happens.

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called [peripheral neuropathy](#). You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Bladder irritation

Cyclophosphamide may irritate your bladder and cause discomfort when you pee (pass urine).

- Drink at least 2 litres (3½ pints) of fluids every day, especially during the 24 hours after chemotherapy.
- Empty your bladder regularly and to try to pee as soon as you feel the need to.
- Contact the hospital straight away if you feel any discomfort or stinging when you pee, or if you notice any blood in your urine.

Effects on the liver and kidneys

This treatment can affect how your kidneys or liver work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys and liver are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids also helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Hearing changes

This treatment may cause hearing changes, including hearing loss. You may have ringing in the ears. This is called tinnitus. You may also become unable to hear some high-pitched sounds. Hearing changes may get better after this treatment ends. But this does not always happen. If you notice any changes in your hearing, tell your doctor, nurse or pharmacist.

Shaking hands (tremor)

Thalidomide might cause your hands to become shaky or to shake more than usual. Let your doctor, nurse or pharmacist know if you notice this. It will get better after your treatment ends.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Effects on the heart

Sometimes thalidomide can affect your heart. It may feel like your heart is beating too slowly. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call 999 if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

Low blood pressure

[Thalidomide](#) can cause a temporary fall in your blood pressure. This can make you feel dizzy for a few moments if you stand up quickly. It can help to move slowly from lying to sitting and from sitting to standing.

Tell your doctor or nurse if you have ever had problems with your blood pressure.

Second cancer

This treatment can increase the risk of developing a second cancer years later. This is rare. The benefits of treatment usually far outweigh this risk. Your doctor, nurse or pharmacist can give you more information.

Hepatitis B and shingles

Some viral infections can become active again while you are taking this treatment. This includes hepatitis B (a liver infection) and herpes zoster (shingles). These may need treatment with anti-viral drugs. Your doctor or nurse will talk to you about this. Before you start treatment, you will have a

blood test to check if you have had hepatitis B.

Other information

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Lactose

This treatment may contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

Donating blood or sperm

You must not give blood or donate sperm during treatment or for 1 week after treatment has finished.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Driving

Do not drive if you have dizziness or blurred vision, or if you feel tired or sleepy. Talk to your doctor if you are not sure whether it is safe for you to drive.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help